

Dataset Description

Synthetic Financial Transaction Dataset

Overview

A synthetic dataset of financial transactions generated using a deterministic pseudo-random generator (seed=42). Designed to simulate realistic banking transaction patterns with embedded fraud cases for ML model training and evaluation.

Data Generation

- **Generator:** TransactionProducer class (backend/pipeline/data_ingestion.py)
- **Generation method:** Linear congruential generator for reproducibility (seed=42)
- **Size:** Configurable (5000-10000+ records), generated on-the-fly
- **Fraud rate:** ~6-10% of transactions are fraudulent
- **Fraud assignment:** A transaction is flagged as fraud if `risk_score > 72` OR random probability < 6%

Schema (16 Fields)

#	Field	Type	Description	Example
1	transaction_id	String	Unique transaction identifier	TXN-0000001
2	timestamp	DateTime	Transaction timestamp (UTC)	2025-01-15T14:30:00
3	amount	Float	Transaction amount in USD (\$0.50 - \$50,000)	1245.67
4	merchant	String	Merchant name (8 possible)	TechMart Pro
5	region	String	Geographic region (6 possible)	NA-EAST
6	channel	String	Transaction channel (5 possible)	WEB, MOBILE, ATM, POS, API
7	fraud_type	String or null	Type of fraud if applicable (6 types)	Card Skimming
8	risk_score	Integer	Risk score 0-99 (higher = riskier)	72
9	status	String	Transaction status	FLAGGED

#	Field	Type	Description	Example
			(5 possible)	
10	user_id	String	User identifier (USR-0001 to USR-9999)	USR-1234
11	is_fraud	Boolean	Ground truth fraud label	True
12	latitude	Float	Latitude coordinate	40.7128
13	longitude	Float	Longitude coordinate	-74.0060
14	device_id	String	Device fingerprint hash	a1b2c3d4
15	ip_address	String	IP address	192.168.1.1
16	processing_ms	Integer	Processing time in milliseconds	45

Fraud Types

Type	Description
Card Skimming	Unauthorized card data capture
Account Takeover	User account compromised
Synthetic ID	Fake identity used for fraud
Money Mule	Illicit fund transfer through accounts
Phishing	Credential theft via deception
CNP Fraud	Card-not-present transaction fraud

Regions

NA - EAST, NA - WEST, EU - WEST, APAC, LATAM, ME - AF

Merchants

TechMart Pro, GlobalPay, SwiftShop, NexCommerce, PayHub, CryptoXchange, LuxuryGoods, BetaPay

Dataset Statistics (Sample: 5000 records)

Metric	Value
Total transactions	5000
Fraud count	~300-500 (6-10%)
Avg transaction amount	~\$25,000

Metric	Value
Min amount	\$0.50
Max amount	\$50,000
Avg risk score	~50

Usage

```
from backend.pipeline.data_ingestion import TransactionProducer

producer = TransactionProducer(seed=42)
transaction = producer.generate() # single transaction
transactions = [producer.generate() for _ in range(1000)] # batch of 1000
```